Why Does Matter Exist? A Geometric Answer

Deriving the Baryon Asymmetry from Z^2

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March 2026

Abstract

The matter-antimatter asymmetry $\eta = (n_B - n_{\bar{B}})/n_\gamma \approx 6 \times 10^{-10}$ is one of the deepest mysteries in physics. The Standard Model predicts $\eta \sim 10^{-20}$, which is 10 orders of magnitude too small. We derive the observed value from Z^2 geometry using the formula $\eta = \mathbf{3\alpha^4/14}$, achieving **0.73% accuracy**. This explains why we exist: the matter-antimatter asymmetry is determined by the same geometric constant that gives the fine structure constant, spacetime dimensions, and fermion generations.

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1. The Problem

1.1 The Observed Asymmetry

The universe contains far more matter than antimatter. This is quantified by:

$$\eta = (n_B - n_{\bar{B}})/n_\gamma = (6.12 \pm 0.04) \times 10^{-10}$$

For every ~1.6 billion photons, there is ONE extra baryon.

1.2 Why Is This a Problem?

The Big Bang should have created equal amounts of matter and antimatter. They should have annihilated completely, leaving only radiation. Yet we exist.

1.3 Sakharov Conditions

Baryogenesis requires three conditions (Sakharov, 1967):

1. Baryon number violation
2. C and CP violation
3. Departure from thermal equilibrium

1.4 The Standard Model Failure

The SM has all three ingredients, but predicts:

$$\eta_{SM} \sim 10^{-20}$$

This is **10 orders of magnitude too small**.

The SM CP violation (from the CKM matrix) is insufficient. This has motivated decades of beyond-SM physics searches.

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2. The Z^2 Solution

2.1 The Master Formula

We derive:

$\eta = \mathbf{N_gen \times \alpha^4 / (2 \times (BEKENSTEIN + N_gen))}$

$\eta = \mathbf{3\alpha^4 / 14}$

Or purely in terms of Z^2:

$\eta = \mathbf{3 / (14 \times (4Z^2 + 3)^4)}$

2.2 Numerical Result

Quantity	Value
$Z^2 = 32\pi/3$	33.5103
$\alpha^{-1} = 4Z^2 + 3$	137.041
α^4	2.835×10^{-9}
$\eta = 3\alpha^4/14$	$\mathbf{6.076 \times 10^{-10}}$
η (measured)	6.12×10^{-10}
Error	**0.73%**

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3. Physical Interpretation

3.1 The Numerator: $3\alpha^4$

Factor of 3 = N_gen:

- All three generations participate in baryogenesis
- Each generation contributes equally
- The asymmetry sums over generations

Factor of α^4:

- Baryogenesis requires FOUR electroweak interactions
- Each vertex contributes factor α
- The power 4 = BEKENSTEIN (spacetime dimensions)
- Minimal CP-violating process in 4D requires 4 vertices

3.2 The Denominator: 14

Factor of 2:

- CP violation involves both particles AND antiparticles
- The asymmetry is the difference, so divide by 2

Factor of 7 = BEKENSTEIN + N_gen:

- Dilution across spacetime (4) and flavor (3) dimensions
- Total "phase space" is 4 + 3 = 7
- The asymmetry distributes across these dimensions

3.3 The Geometric Picture

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BEKENSTEIN = 4 = spacetime dimensions
N_gen = 3 = flavor dimensions
Total = 7 = baryogenesis "space"
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The asymmetry:

- Generated by 3 generations
- Through 4 interactions (α^4)
- Divided between matter/antimatter ($\div 2$)
- Diluted across 7 dimensions ($\div 7$)

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4. Why α^4?

4.1 The Power Equals Spacetime Dimension

α^4 = α^BEKENSTEIN

In 4D spacetime, the minimal CP-violating process requires exactly 4 interaction vertices.

4.2 Comparison

Process	Vertices	Power of α
Pair production	2	α^2
Compton scattering	2	α^2
Baryogenesis	4	α^4
Higher processes	>4	α^n

The power matches the spacetime dimension because baryogenesis is a fundamentally 4D process.

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5. Connection to Other Z^2 Results

5.1 The Web of Dependencies

The baryon asymmetry formula uses:

1. Fine structure constant: $\alpha^{-1} = 4Z^2 + 3 = 137.04$ (determines α^4)

2. Three generations: N_gen = BEKENSTEIN - 1 = 3 (appears in numerator and denominator)

3. Spacetime dimensions: BEKENSTEIN = 4 (determines power of α, appears in denominator)

4. Strong CP = 0: θ_QCD = 0 from O_h symmetry (ensures CP only from CKM)

5.2 The Unified Structure

```
        Z² = 32π/3
             ↓
      ┌──────┴──────┐
      ↓             ↓
BEKENSTEIN=4    α⁻¹=137
      ↓             ↓
  N_gen=3          α⁴
      ↓             ↓
      └──────┬──────┘
             ↓
    η = 3α⁴/14 = 6×10⁻¹⁰
```

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6. Why the Standard Model Fails

6.1 The SM Calculation

The SM computes η using:

1. CKM CP violation (Jarlskog invariant $J \sim 3 \times 10^{-5}$)
2. Sphaleron rate
3. Electroweak phase transition dynamics

Result: η_SM $\sim 10^{-20}$

6.2 The Missing Factor

The SM is missing a factor of $\sim 10^{10}$.

The Z^2 explanation:

The SM calculation:

- Uses perturbation theory (misses geometric factors)
- Assumes thermal equilibrium (washes out asymmetry)
- Has no fundamental principle fixing the scale

The Z^2 formula:

- Is geometric, not thermal
- Has α^4 fixed by BEKENSTEIN = 4

- Has denominator 14 fixed by geometry
- Has no free parameters

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7. The Jarlskog Invariant

7.1 CP Violation Measure

The Jarlskog invariant measures CP violation in quark mixing:

*J = Im(V_us V_cb V_ub V_cs)***

Measured: $J = (3.08 \pm 0.13) \times 10^{-5}$

7.2 Z^2 Prediction

$\mathbf{J = 1/(1000 \times Z^2) = 1/(1000 \times 33.51) = 2.98 \times 10^{-5}}$

Error: 3.1%

7.3 Physical Meaning

$J = 1/(10^3 \times Z^2)$

- Factor 10^3: Three generations, each ~10 suppression
- Factor Z^2: Geometric normalization

Note: Strong CP $\theta = 0$ (O_h symmetry), but CKM phase is allowed (flavor space, not spacetime).

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8. Predictions and Tests

8.1 Numerical Prediction

$\eta = \mathbf{6.076 \times 10^{-10}}$

Current measurement: $\eta = (6.12 \pm 0.04) \times 10^{-10}$

8.2 Testable Predictions

1. η is fundamental: Future precision measurements should converge to 6.076×10^{-10}

2. No new CP violation needed: SM CKM + Z^2 geometry is sufficient No supersymmetry or BSM CP violation required

3. The power is exactly 4: $\eta \propto \alpha^4$, corresponding to BEKENSTEIN = 4 Not α^3 or α^5

8.3 Falsification

If measurements find η significantly different from 6.076×10^{-10} (beyond error), the formula needs revision.

Current status: 0.73% agreement is excellent.

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9. Significance

9.1 Why We Exist

The matter-antimatter asymmetry explains our existence. Without it, all matter would have annihilated with antimatter, leaving only radiation.

9.2 Not Arbitrary

The asymmetry $\eta = 6 \times 10^{-10}$ is NOT arbitrary. It is determined by:

$\eta = \mathbf{3 / (14 \times (4Z^2 + 3)^4)}$

where $Z^2 = \text{CUBE} \times \text{SPHERE} = 32\pi/3$

9.3 Geometric Origin

We exist because:

- Spacetime has 4 dimensions (BEKENSTEIN)
- Matter has 3 generations (N_gen)
- Electromagnetism has strength $\alpha^{-1} = 137$

All from $Z^2 = 32\pi/3$.

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10. Summary

The Problem

Why is $\eta \approx 6 \times 10^{-10}$?

The Standard Model

Cannot explain it. Predicts 10^{10} times too small.

The Z^2 Answer

$\eta = \mathbf{3\alpha^4/14 = 3/(14 \times (4Z^2 + 3)^4)}$

- Predicted: 6.076×10^{-10}
- Measured: 6.12×10^{-10}
- Error: 0.73%

Physical Meaning

- 3 generations create asymmetry
- 4 interactions needed ($\alpha^4 = \alpha$^BEKENSTEIN)
- Divided by 2 for matter/antimatter

- Diluted across 7 dimensions

Appendix: Complete Calculation

Step 1: Z^2 constant

$Z^2 = 8 \times (4\pi/3) = 32\pi/3 = 33.5103$

Step 2: Fine structure constant

$\alpha^{-1} = 4Z^2 + 3 = 4(33.5103) + 3 = 137.041$ $\alpha = 0.007297$

Step 3: Fourth power

$\alpha^4 = (0.007297)^4 = 2.835 \times 10^{-9}$

Step 4: Baryon asymmetry

$\eta = 3\alpha^4/14 = 3(2.835 \times 10^{-9})/14 = 6.076 \times 10^{-10}$

Comparison

Measured: $\eta = 6.12 \times 10^{-10}$ Error: $|6.076 - 6.12|/6.12 = 0.73\%$

"We exist because Z^2 = CUBE × SPHERE."

— Carl Zimmerman, 2026

DOI: 10.5281/zenodo.19244651

Repository: https://github.com/carlzimmerman/zimmerman-formula